

FROM THE SPREAD OF IR TO THE SECOND IR

ALFANI-COLLI, CH. 5.3 AND 6

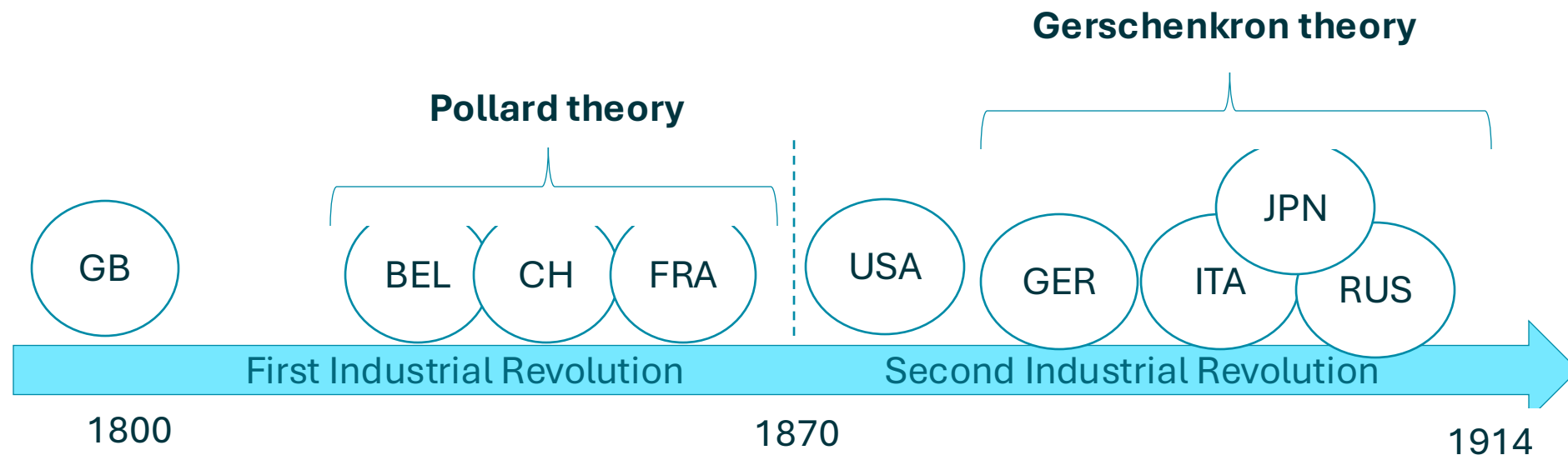
Economic history

A.y. 2025/2026

Sonia Schifano

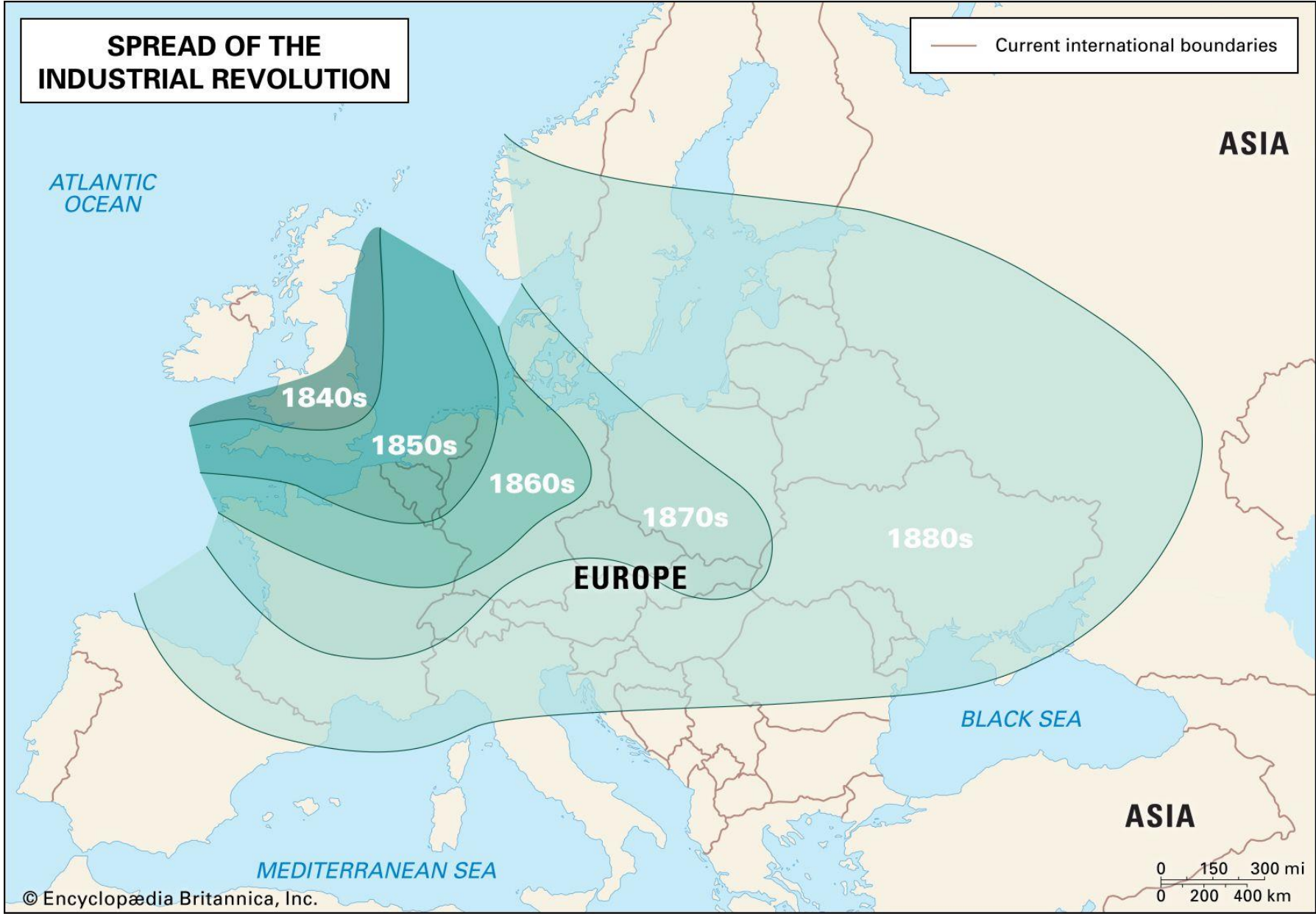
IN THIS CLASS

- When and how industrialisation spreads (regional factors)
- When industry does not take off on its own
- The role of universal banks
- The role of the State



THE LONG WAY TO INDUSTRIALISATION

- England strenuously defended 'its' inventions but:
 - from 1824 English craftsmen could legally expatriate and from 1825 they could start producing machinery abroad under licence.
 - from 1843, with the adoption of a free-trade policy, restrictions on the movement of labour were completely removed.
- This favoured the spread of the new production systems across the whole European continent.



THE LONG WAY TO INDUSTRIALISATION

- The first regions to adopt English technologies were Belgium, Alsace, Rhineland, Northern France, (Protestant) Switzerland, Saxony, Bohemia, and Lombardy.
- **Why? They were already part of mercantile exchange circuits and/or had a well-established manufacturing tradition.**
- These regions began by imitating the production and the organisational system of the English industry, looking in particular at the textile sector.

POLLARD THEORY

- According to a model theorised by Sidney Pollard, the spread of industrialisation in Europe is the result of numerous **regional developments**:
 - development does not take place nationally, but regionally (i.e. by homogeneous areas, not necessarily belonging to the same country)
 - homogeneity in resources and human capital was particularly crucial
 - all countries have developed areas and backward areas

! Given the same resources, the institutional contexts were often very different and therefore development had different speeds and characteristics from region to region!

POLLARD THEORY

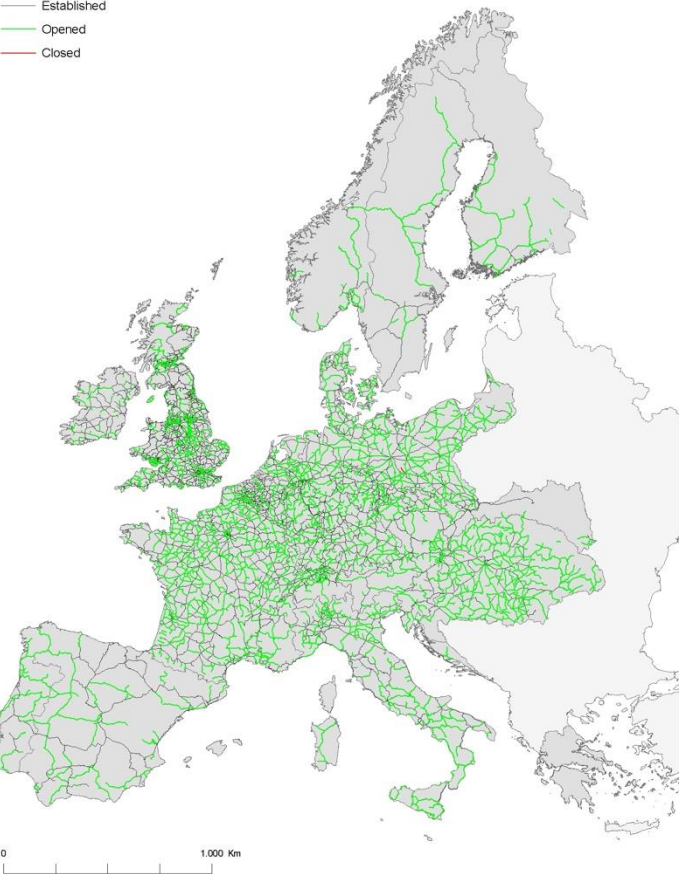
- ▶ An additional concept introduced by Pollard and important for understanding industrialization is the so-called "**contemporaneity differential**":
- ▶ Developments in the international economy interfere with the development plans of individual countries, directing their effects either positively or negatively.
- ▶ An example is the railways, built in all European countries and in the USA during the 19th century, but with very different results:
 - Great Britain = followed industrial development and were a consequence of it.
 - France, Germany, USA = the railways promoted development.
 - Italy = the railways caused great expenses without particular benefits.



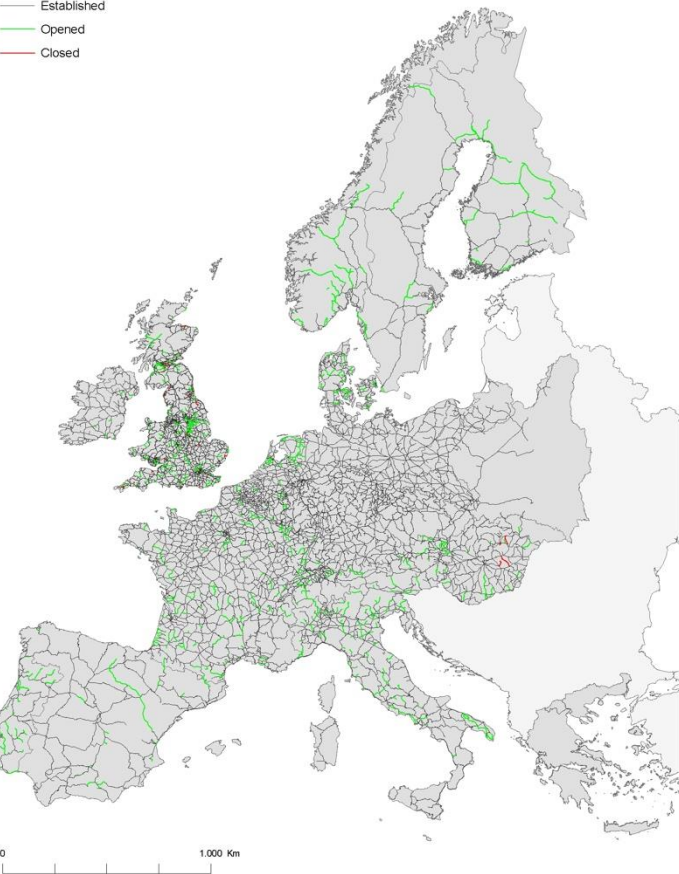
Railways network in Europe (1870)



Railways network in Europe (1870 - 1900)



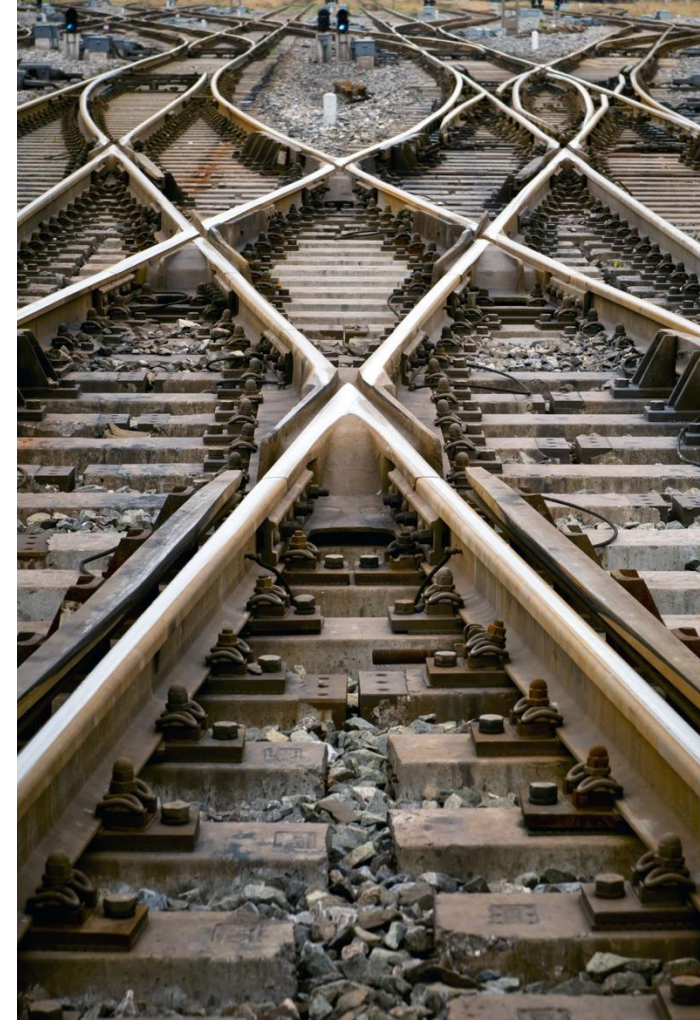
Railways network in Europe (1900 - 1930)



RAILWAYS AND ECONOMIC GROWTH

► WHY ARE RAILWAYS SO IMPORTANT?

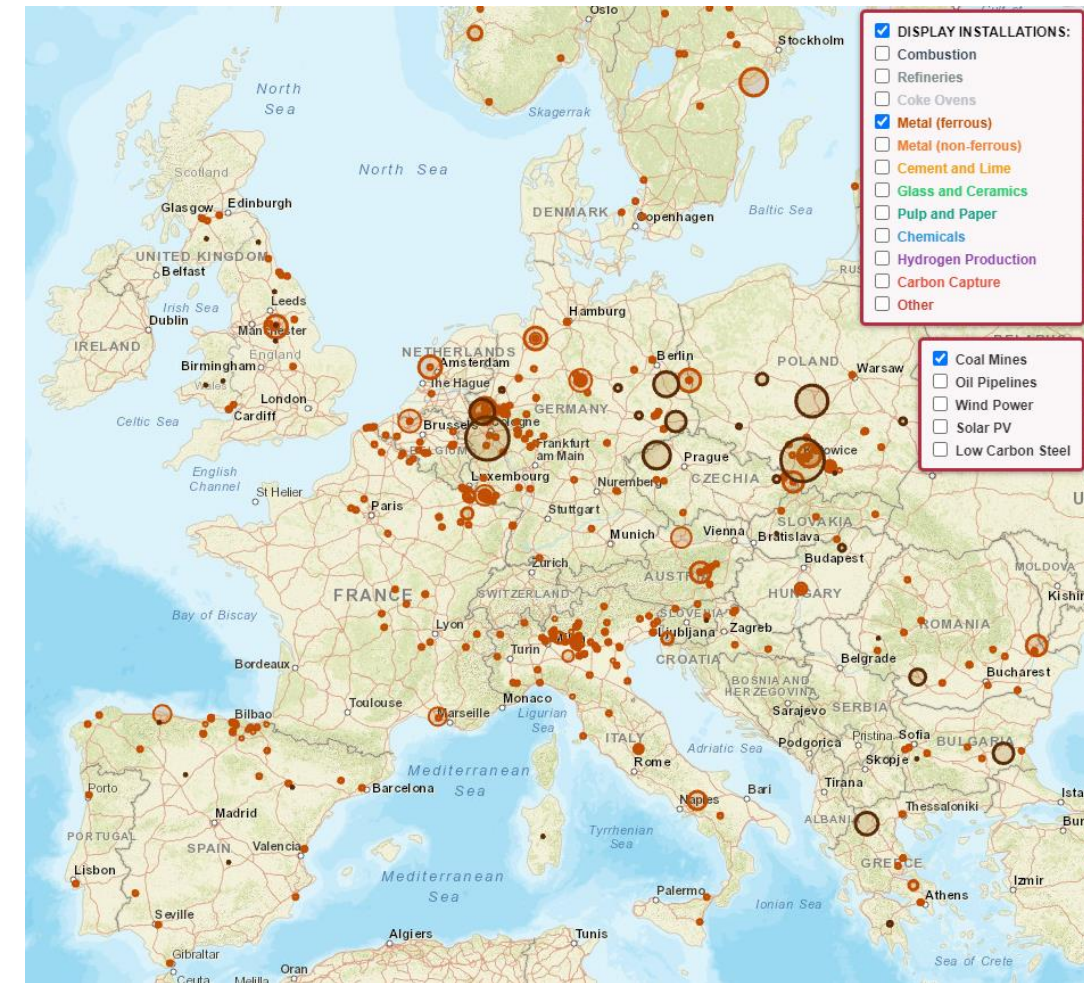
1. **Economic and industrial impact:** The railway system was a powerful engine for economic growth. It created a massive demand for materials like coal, iron, and steel, which in turn stimulated the growth of industries like mining, metallurgy, and engineering. The construction of railways themselves became a major industry, employing millions and spurring innovation in civil engineering.
2. **Social and cultural transformation:** For the first time, mass travel became accessible to the general public. This facilitated migration from rural areas to burgeoning urban centers, fueling the growth of cities and suburbs. The railway also accelerated the spread of ideas and information.
3. **Global expansion and colonialism:** The railway revolution was not confined to Europe and North America. It became a crucial tool for imperial expansion, particularly in colonies like India and across Africa. Colonial powers used railways to efficiently transport raw materials from the interior to coastal ports for export.



BELGIUM

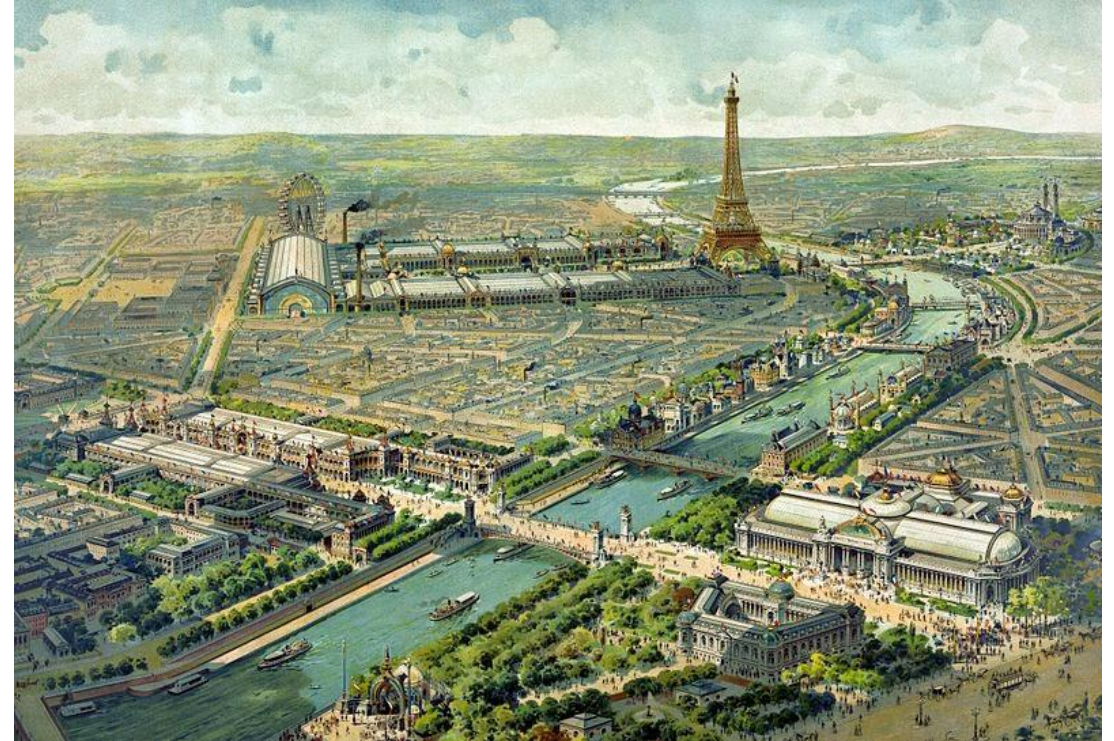
Was the first country to adopt the British model

- Institutions
- Proximity
- Industrial tradition – manufacture – cotton - linen
- Commerce
- Transportation – State intervention in 1830 for a basic railway network
 - Infusion of foreign technologies
- Natural resources
 - Mining industry
 - Canals connecting Belgium with northern France
 - Attractive incentives from French capital



FRANCE-THE FRENCH PARADOX

- Backwardness (but different phases) 1700-1750-1800-1850
- Low demographic growth
- French Revolution
- Napoleonic Civil Code and Commercial Code
- No richness in coal – importing 30% of coal supplies
- Machine production – export and domestic textile industry
- Small scale enterprises – water dependence
- Food self-sufficiency



THE SECOND INDUSTRIAL REVOLUTION



THE DOUBLE LEAP

- ▶ Late-industrializing nations faced a unique «double leap».
- ▶ This dual challenge required massive capital and coordinated strategy, often beyond the capacity of the private sector alone.
- ▶ The two fronts:
 - ▶ **First Industrial Revolution (Catch-up):** They had to modernize existing industries, particularly in areas like textiles, coal mining, and iron production. These were the foundational sectors of the initial industrial age.
 - ▶ **Second Industrial Revolution (New Frontiers):** Simultaneously, they had to invest heavily in the emerging, capital-intensive sectors of the new era.



THE PILLARS OF THE SECOND INDUSTRIAL REVOLUTION

- Scientific knowledge applied to production processes: more and more **R&D**.
- Emergence of big business with **new organisational structures**.
- New industrial sectors → **increasingly capital intensive**
- The United States outperforms Great Britain to become the centre of the world economy.

THE PILLARS OF THE SECOND INDUSTRIAL REVOLUTION

The **new technologies** have different characteristics to those of the first industrial revolution:

- **greater capital intensity**
- much closer relationship between **science and technology**
- greater number of **industrial sectors** involved in innovation

Within a few decades, industrial **R&D** laboratories become key locations for technological innovation processes and science becomes key to generating innovation

Growth accelerates at a faster pace than during the first industrial revolution.

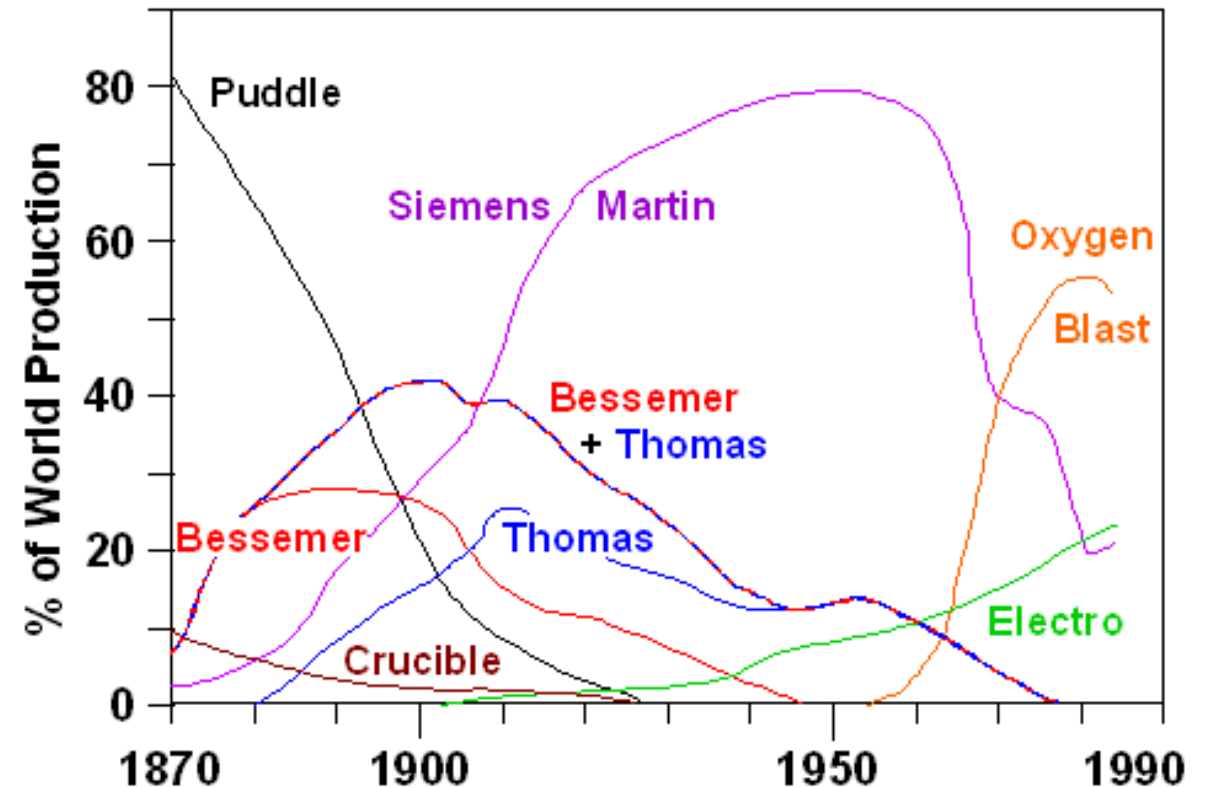
INDUSTRIALISATION AND ECONOMIC GROWTH

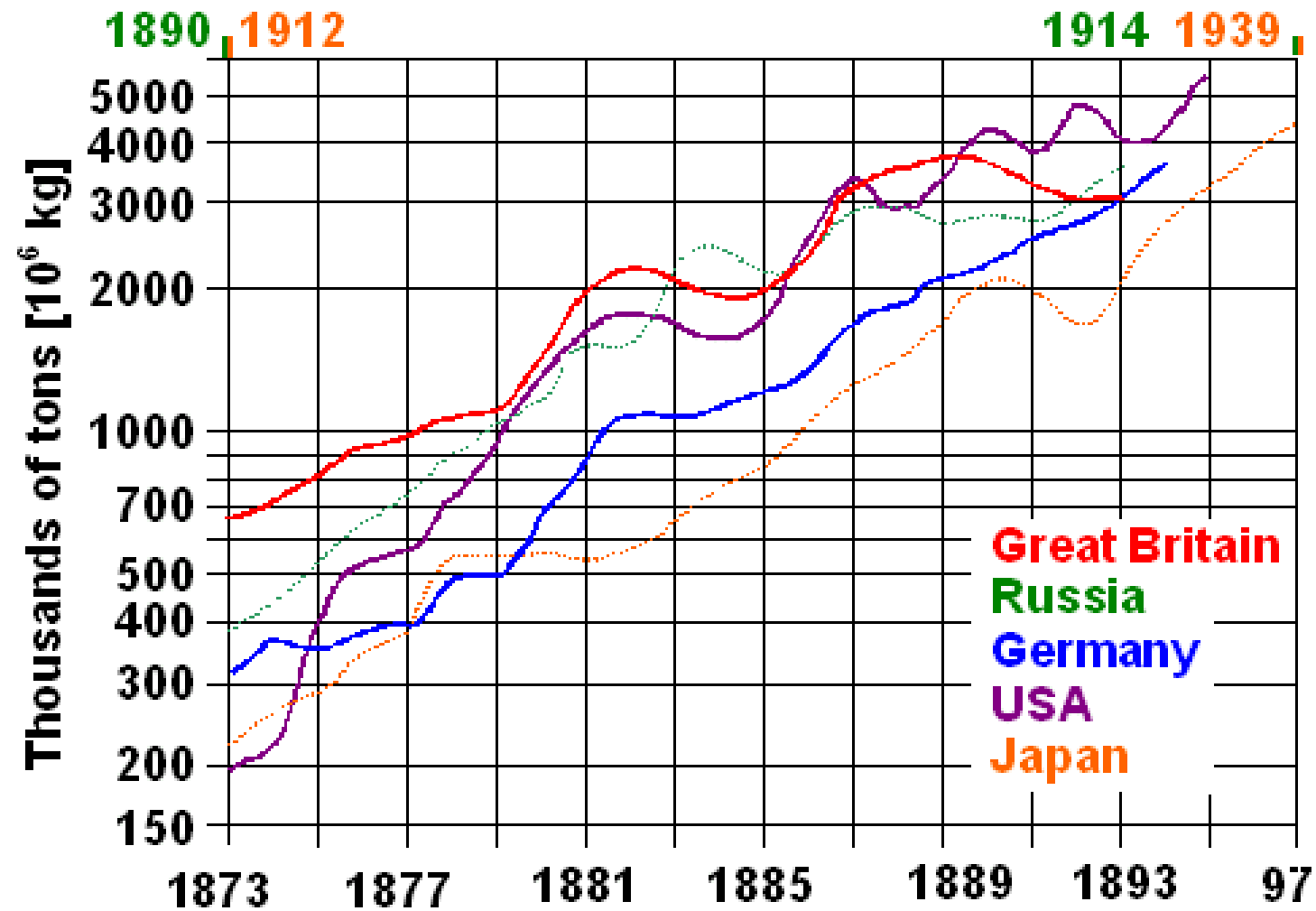
Table 2–22b. **Growth of Per Capita GDP in European Colonial Powers and Former Colonies, 1500–1998**
(annual average compound growth rates)

	1500–1700	1700–1820	1820–1913	1913–50	1950–98
Britain ^a	0.31	0.34	0.96	0.80	2.10
France	0.15	0.18	1.13	1.12	2.77
Italy	0.00	0.01	0.90	0.85	3.44
Netherlands	0.52	–0.12	0.86	1.07	2.56
Portugal	0.15	0.10	0.27	1.38	3.89
Spain	0.13	0.14	0.81	0.17	3.78
China	0.00	0.00	–0.08	–0.62	4.17
India	0.00	–0.03	0.25	–0.23	2.18
Indonesia	0.01	0.04	0.42	–0.20	2.74
Brazil	0.17	0.12	0.24	1.97	2.50
Mexico	0.15	0.24	0.89	0.85	2.18
United States	0.14	0.73	1.56	1.61	2.21
Ireland ^b	0.15	0.17	1.23	0.63	3.53

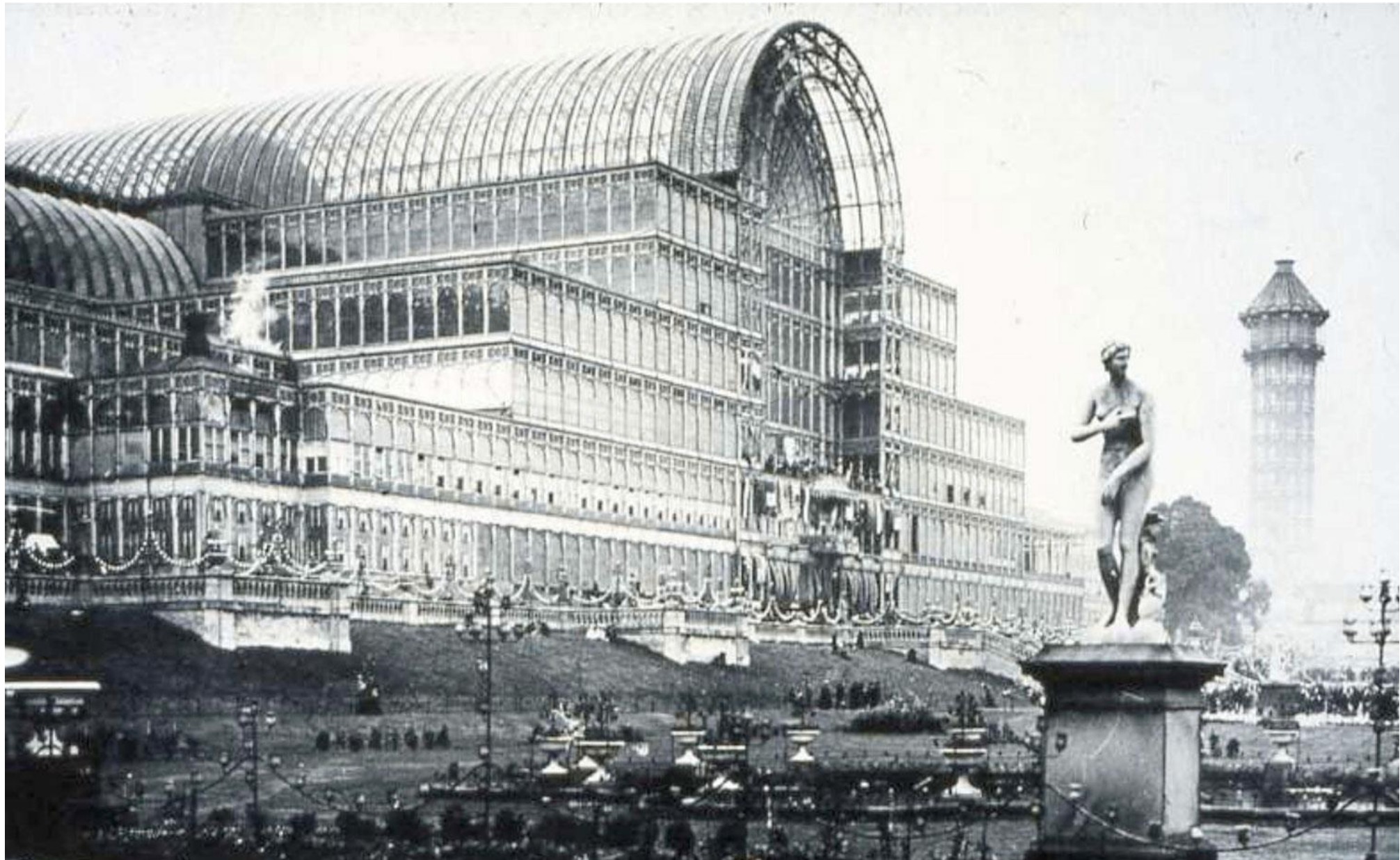
THE PILLARS OF THE SECOND INDUSTRIAL REVOLUTION

- **Steel production:** new processes invented between 1856 and 1876 allow mass production. Volumes of output soar from the 1880s.
- 1856: Bessemer method
- 1856-67: Martin-Siemens oven
- 1878-79: Gilchrist Thomas method





*Early iron / steel production in some key countries.
Note logarithmic scale and change of time scale for Russia and Japan*





THE PILLARS OF THE SECOND INDUSTRIAL REVOLUTION

Also fundamental was the development of the chemical industry:

- artificial dyes
- fertilisers
- pharmaceutical products
- explosives (nitro-glycerine)
- synthetic fibres

Organic chemistry is the field in which Germany acquires lasting international leadership thanks to effective cooperation between **universities and industry**

→ **Science!**

THE PILLARS OF THE SECOND INDUSTRIAL REVOLUTION

Electricity

- electric motor and dynamo (1820s-30s)
- telegraph (1840s-60s)
- electricity generation and distribution (1880s)

Food processing/preservation processes

- canning, used especially in the USA since the 1860s
- mechanical refrigeration

Internal combustion engine

- automobile (Daimler & Benz; Diesel, 1880s-90s)

Light mechanics

- typewriter
- sewing machine

THE PILLARS OF THE SECOND INDUSTRIAL REVOLUTION

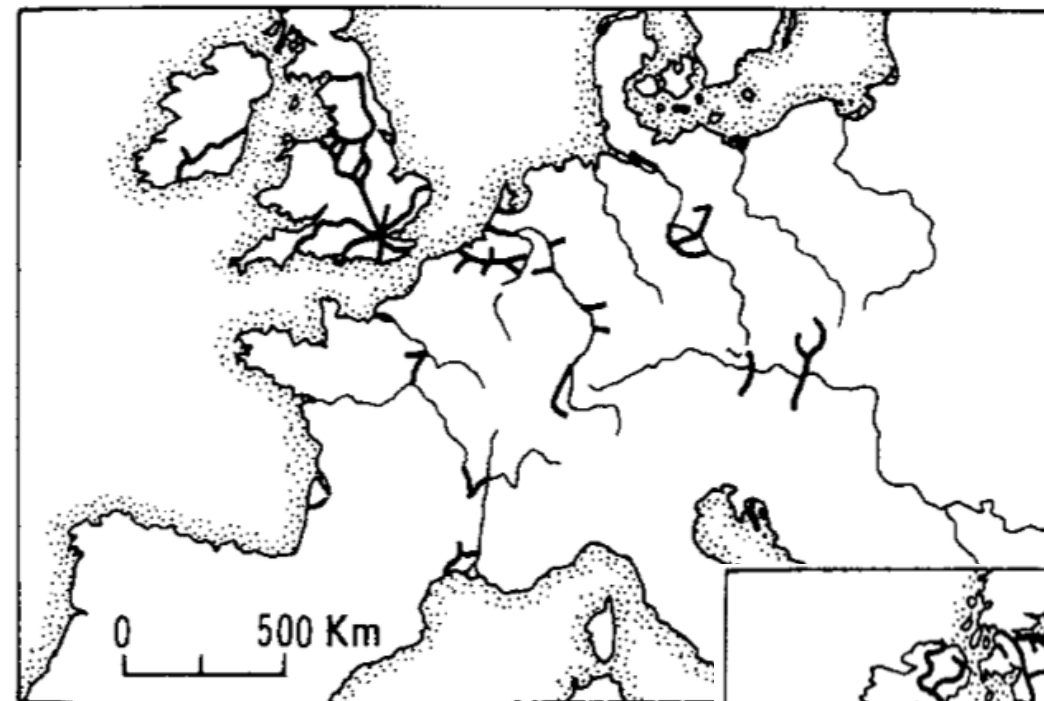
The transition from the factory of the first industrial revolution to the large modern enterprise of the late 19th century is facilitated by new technologies and by the enlargement of markets.

The mid-19th century saw a revolution in transport and communications:

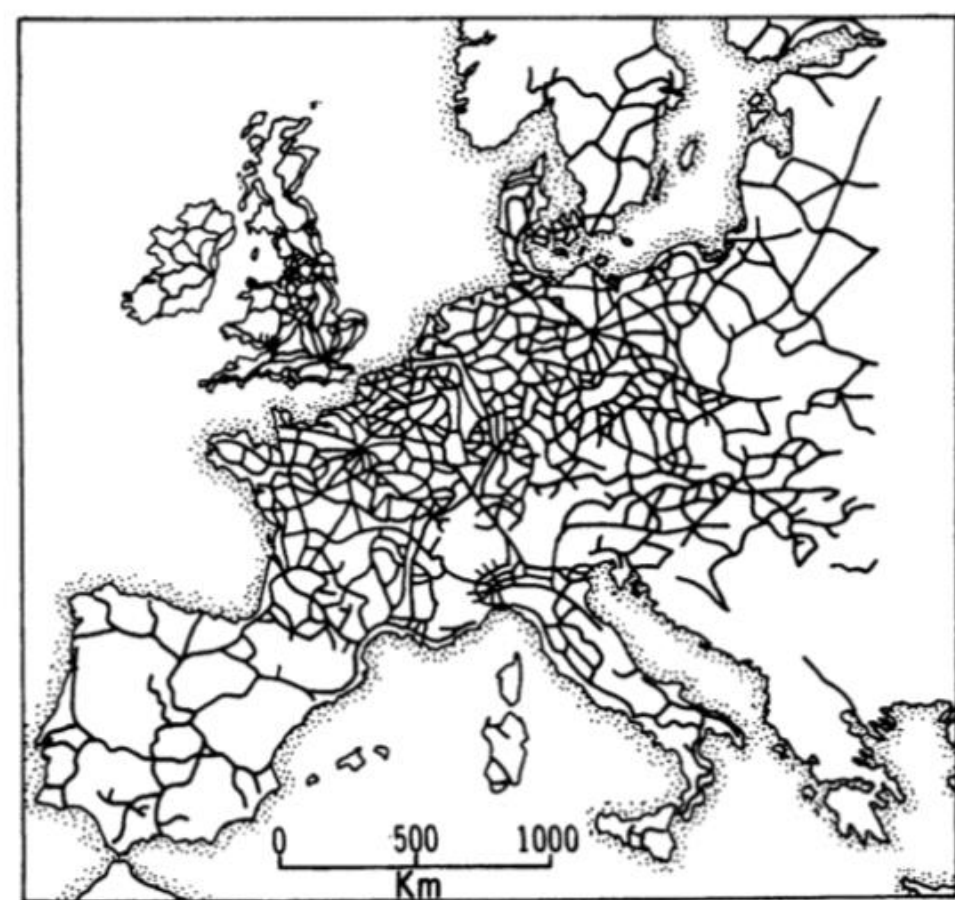
- telegraph (1840), transatlantic cables (1860), telephone (1870)
- trains and steam navigation
- Suez Canal (1859-69) and Panama Canal (1881)

The main consequences were:

- **the decline in transport costs**
- the **creation of a national (and later international) market** served quickly, extensively and reliably

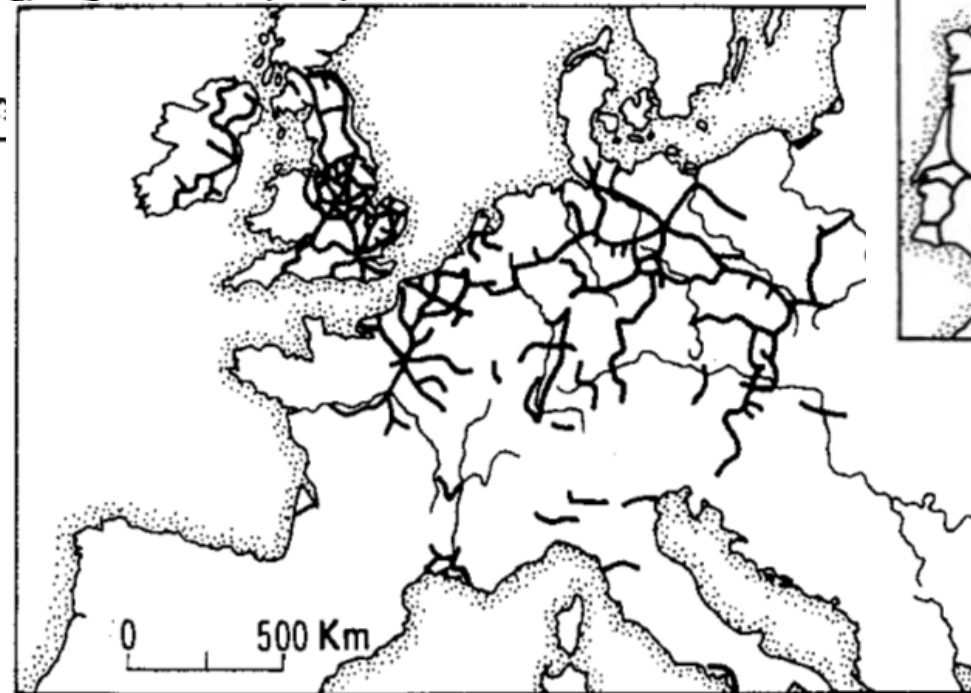


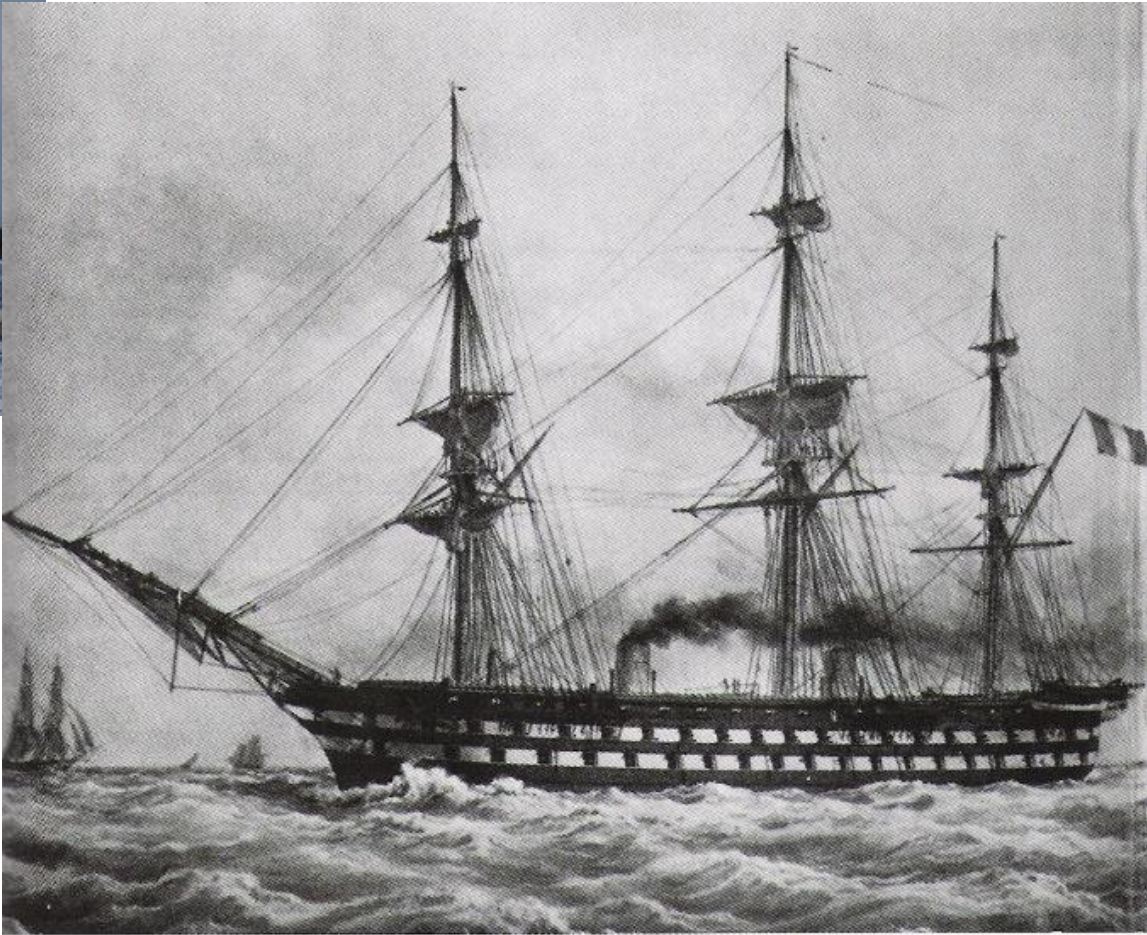
1840



1880

1850





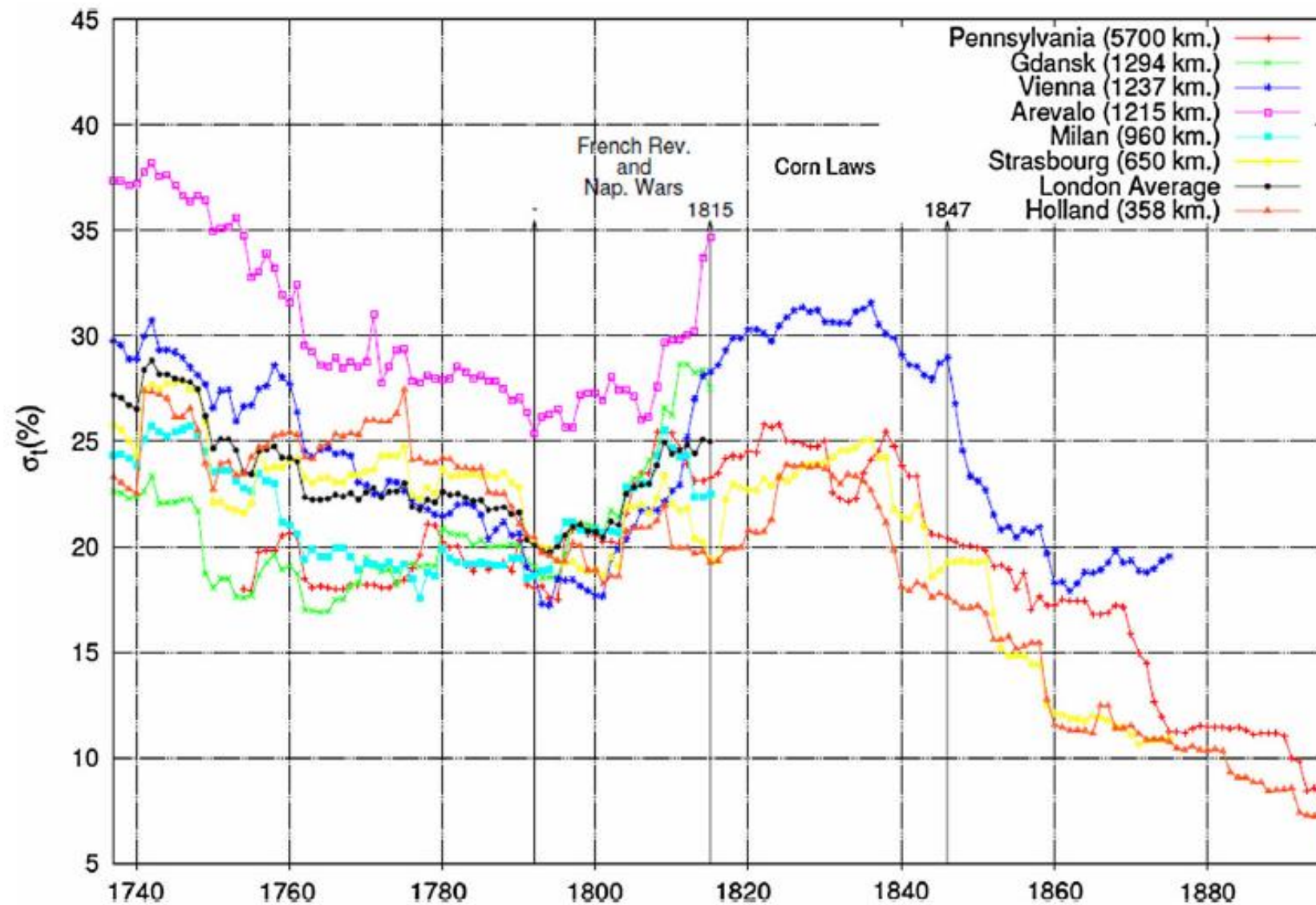


Fig. 1 Market integration in the West (London as *numéraire*), 1737–1896. *Source* Dobado et al. (2012)

THE RISE OF BIG BUSINESS

Constitutive characteristics of the large enterprise:

- facilities capable of achieving **economies of scale**
- **vertical integration**: raw materials → intermediate products → distribution
- **managerial hierarchies** (organisational evolution)

The emergence of the large enterprise alters the dynamics of competition:

- strong industrial concentration → **oligopolistic markets**
- penetration of foreign markets → ***multinationalisation***

...and generates dualism between sectors at the technological frontier and sectors with traditional technologies

THE RISE OF BIG BUSINESS

Big business brings with it major changes in the organisation of work.

In the Victorian factory

→ the organisation of production (time and methods of work) is in the hands of **department heads**.

In the factory system that emerged between the 19th and 20th centuries

→ the system of organisation is in the hands of **managerial figures**, members of the management team.

The main innovators in this field are Frederick Taylor, Henry Ford, Charles Eugène Bedaux, and Taiichi Ohno.

HOW CAN BACKWARDNESS BE ADDRESSED?

THE GERSCHENKRON MODEL

- **Alexander Gerschenkron** described **the advantages of backwardness**. In his model, it is not the long-term dynamics that are central, but above all the phases of '**preparation**' and economic take-off.
- Since each nation has a unique **pre-set of conditions** that differ from those of its neighbours, it goes without saying that the various national development processes will each be different.
- The more distant prerequisites a country has from the leading economy, the more it will have to change its development model in order to be successful. Conversely, the more a country has similar prerequisites to the first comer, the more the model will be able to follow the one adopted by the latter.

THE GERSCHENKRON MODEL

- In this model, backwardness brings advantages: it will be possible to imitate the winning technologies of the other competitors without going through a process of slow research and development.
- In this way there will be:
 - faster development
 - fast-pace development of large-scale industry
 - less agricultural growth
 - greater importance of institutions
 - greater dependence on foreign capital and resources

THE GERSCHENKRON MODEL

- The ‘natural’ consequence of this model is that the state/region that initially holds the leadership is not necessarily the same one that continues to prosper.



- England vs US

(we are applying this model to industrialisation, but it is also valid within a single industrial sector...)

THE LONG WAY TO INDUSTRIALISATION

- Being at a disadvantage also allows backward countries to find substitutes for the factors necessary for industrialization lacking in their own countries, just as the advanced countries did in the past.
- Gerschenkron in his essay *Economic Backwardness in Historical Perspective* (published in 1962) called them **substitute factors**.
- The state and the universal bank system, for instance, played a central role in the development of many European countries on the Continent.

THE ROLE OF SUBSTITUTIVE FACTORS: UNIVERSAL BANKS

As we have seen, economic growth generates the need for a supportive and more articulated financial system. In England the banking system had time not only to develop, but also to better articulate and to (hyper) specialise itself, creating a number of different services able to cover the different needs of the market.

In the **Continent**, the much faster pace of growth led to the creation of **Universal Banks**: institutes more 'generalist' in nature that could offer a number of financial services, including **both commercial and investment banking** services.

Universal Banks are **joint-stock companies** and their specificity makes them highly exposed to the risk of liquidity crises.

THE ROLE OF SUBSTITUTIVE FACTORS: THE STATE

After the mid-19th century, the conditions for industrial development became increasingly difficult due to the high costs of production plants and due to the increasing relative backwardness of the laggard countries.

The role of the state in promoting industrialisation processes became decisive and was expressed through a range of instruments:

- protectionist policies
- fiscal system
- state contracts
- subsidies
- financing and management of infrastructure and enterprises
- industrial bailouts / rescue

GERMANY



- Up to 1833 – awareness of the economic changes occurring in France, Belgium, Britain
- From 1834 to 1870 imitation and borrowing
 - Modern industry (transportation – finance)
- After 1870 industrial supremacy in Western Europe
- Its supremacy was based on production of capital goods or intermediate products for industrial production
 - Coal, iron, steel
 - Chemicals, electricity
- Rapid urbanisation
- Large size of firms
- Cartels+protectionists policies in 1879 = high prices in the domestic market + exports (at competitive prices)

ITALY

The late national unification created a pressing need for economic standardization!

1870s–1890s:

- Late 1870s: some results in textiles (roots in preindustrial period) and railways
- Increasingly high customs tariffs
- A first phase of industrial strengthening in cotton, metallurgy, mechanical and chemical sectors
- Decisive role of the two major mixed banks (COMIT and CREDIT) in supporting industry
- From 1896 onward: a turning point, thanks in part to hydroelectric power and related industries. Large enterprises in innovative sectors (Fiat), growth in textiles, agri-food, metallurgy, and chemicals
- Uneven development: the “Industrial Triangle” (Milan–Turin–Genoa)
- Economic growth but also challenges. Mass emigration (painful!), brought significant benefits (demographic pressure, remittances, and export networks abroad)



TAYLORISM OR *SCIENTIFIC MANAGEMENT*

Mechanical engineer and business consultant, Frederick Taylor is the founding father of the scientific organisation of work (*Principles of scientific management*, 1911)

The elements that characterise it are:

- identification of the best way of doing things
- personnel selection and training
- division of labour and management tasks

Taylor's goal is to increase the **efficiency** of the production process in order to **increase profit**.

FORDISM

The Fordist method overcomes the selection and training problems of scientific management: it does not “*teach the worker the only best way to do his job*”, but “*arrange things in such a way that he can only do his best work*”.

It maximises the principle of **standardisation and economies of scale**:

- The tasks of individual workers are parcelled out
- movements reduced to a minimum
- the required skills are practically non-existent

The (engineering) design of the production layout and the assembly line solve the problem of work organisation.

To this Ford combined

- high wages,
- corporate welfare, and
- total intolerance towards any form of trade union.

